§4 TIPA, PSL and ISL — part 1 of 4

Reference: sec:tipa, subsec:s4_scope, subsec:tipa_architecture, subsec:m1

Source: paper/section4_tipa.tex, lines 1–876 (opening of §4, §4.1 scope and common conventions, §4.2 TIPA and the mechanism M1), at commit e6e12cd. The Japanese manuscript is authoritative. Part 1 of §4 follows 03_principles_5.md. Part 2 (M2 and M3, lines 877–1683) follows in 04_tipa_2.md, part 3 (PSL and the description of purpose, action and result, lines 1684–2349) in 04_tipa_3.md, and part 4 (ISL, integrated high-speed processing and Automaticity, the functional axis, and the closing summary, lines 2350–2964) in 04_tipa_4.md.

Opening of §4

Reference: sec:tipa

§3 formulated SM, EU and FR as constraint axes on an ACS in existence. §4 does not redefine them. For a single ACS at one time, it gives standard functional forms by which information is acquired and integrated, differences relevant to feasibility are evaluated, and responses are formed and connected to effective action.

The standard functional forms treated are:

- **TIPA** (Triadic Information-Processing Architecture), which bears information integration, adaptive evaluation and response-action formation;
- **PSL** (Physical Substrate Layer), which connects TIPA with the implementation side in input, output, substrate maintenance and local protective response;
- **ISL** (Implementation Substrate Layer), which organizes the implementation carriers that can bear sensing, acting, information retention and others.

The responsibility of §4 is the structure, information dependence, form of operation and connections *at the same time*. Memory update, learning, internal-model update, the formation, strengthening, weakening and release of integrated high-speed processing and of Automaticity, changes in the health of PSL and ISL, and other changes from t to $t+1$ are left to §5 (sec:extended_dynamics).

Principle level and mechanism description (rem:principle_level_locus). SM, EU and FR are not identified with any one of TIPA, PSL, ISL. A principle itself is neither satisfied nor unsatisfied; what has a truth value is its time-indexed satisfaction predicate (rem:s3_description_levels). This paper does not state, as a general consequence, the truth or falsity of a time-indexed satisfaction predicate from the mechanism outputs or operating states of §4 alone.

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§4.1 Scope and common conventions

Reference: subsec:s4_scope

This subsection confirms only the responsibility boundaries and the common mathematical devices of §4. Symbols newly used in §4 are listed in app:symbols_s4. Assumptions and distinctions specific to a mechanism are given where that mechanism is treated.

Same-time structure and the boundary with inter-time dynamics (subsubsec:s4_same_time_boundary). §4 treats structure, information dependence, operation and connection of TIPA, PSL and ISL at one time: which information becomes available from which mechanism, what each mechanism generates, and how the action entering the state transition is constructed. Formation, learning, update, degradation, recovery and other changes between times belong to §5.

Constitutively necessary functions versus standard functional forms (subsubsec:s4_abc_application). The classes A, B, C follow def:abc_classification. §4 confirms only that constitutively necessary functions are not identified with TIPA, PSL, ISL and other classification-B forms, and that a classification-C quantity is not regarded as an additional component of an ACS. Concrete organs, circuits, hardware and software modules are not required to corre-

spond one-to-one to a classification-B form. The convention on levels of description (`subsec:description_levels_global`) applies as it is: introducing a formal quantity does not mean the ACS knows its true value, and saying that a structural function is needed does not mean the ACS implements a particular formula of this paper internally. What §4 requires about the existence of a structural function is limited to this: for at least some of the differences related to that function, an evaluation result or an internal response can change.

Since §4 treats a single ACS Σ_i, the individual index is omitted as in §2, writing x_t, e_t, A_t; the collapse indicator, which concerns individual identity, keeps its index, $\mathrm{Col}_{i,t}$.

Deterministic selection projection (`subsubsec:selection_projection`).

Definition (`def:sel_proj`): deterministic selection projection (classification B). For a family $\mathfrak{C}_Y \subseteq \mathcal{P}(Y) \setminus \{\emptyset\}$ of non-empty candidate sets over a set Y, a deterministic selection projection $\mathrm{Proj}_{\mathrm{sel},Y} : \mathfrak{C}_Y \to Y$ is a fixed selection rule with $\mathrm{Proj}_{\mathrm{sel},Y}(C) \in C$ for every $C \in \mathfrak{C}_Y$. It is a common formal device for keeping the determinism of Phase 1 when several equivalent candidates exist. The concrete tie-break rule depends on the implementation.

State carrier and information provenance (`def:s4_carrier_contract`).

Definition (`def:s4_carrier_contract`): carrier convention for same-time operation (classification B). The time-dependent mechanism outputs of §4 are generated, under fixed rules, from (M_t, I_t), where $I_t = P_{\mathrm{sens}}(x_t, e_t)$ and $I_{\mathrm{sense},t} = I_t$. Physical signals, the PSL substrate, sensitivity and effector states are carried by x_t; retained contents are carried by the learning component and working component $M_t^{\mathrm{L}}, M_t^{\mathrm{W}}$. Each signal space and information space is non-empty. Results using a topology use only the topologies of the already defined X and $\mathcal{M}$; compactness of signal spaces and continuity of readouts are not required.

Each previous-time input is given by a fixed readout $\rho_{\mathrm{past}} : \mathcal{M}^{\mathrm{W}} \to \mathcal{P}_{\mathrm{prio}} \times U \times X^{\leq W} \times \mathsf{S}_{M1} \times \mathcal{J}^{\leq W_{\mathrm{short}}} \times \mathcal{P}_{\mathrm{fin}}(\mathsf{Pur}) \times \mathcal{G}_O$. Its components are used as: the retained previous Priority; the record of the effective action; the state-estimate history; the record of M1 output; the sensed-information history; the record of the set of purposes at the previous time; and the record of the external-side prediction set for the previous effective action. The last two records are formed only to the extent that the detailed type of Update (`rem:update_type_two_layers`) takes the intermediate values of the same step as additional arguments. The record of the set of purposes is carried by the working component $\mathcal{M}_3^{\mathrm{W}}$.

Hereafter a previous-time symbol in a mechanism input denotes this recorded value; identity with the actual past output is not assumed. The initial record is also read from M_0^{W}, and no independent initial state is added. Missing or wrong records are permitted, but each mechanism, including its handling of them, is given as a fixed total map. An empty history or a fixed initial convention value is not taken as an actual past value. Where identification of a missing record is needed, that information is held in an existing working component. Missing history is not completed with true past values.

The existence of ρ_{past} is not made a theorem of arbitrary encodability; only records realized under the adopted component types of Update are used. In particular, Update_1 takes no state of another mechanism as a direct input, so this paper does not state that records originating in other mechanisms can in general be stored exactly in mechanism 1. Taking $\mathcal{M}^{\mathrm{W}}$ as a whole as the carrier of the readout does not lift this update constraint.

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§4.2 TIPA: the Triadic Information-Processing Architecture

`Reference: subsec:tipa_architecture`

TIPA := (M_1, M_2, M_3) is a classification-B standard functional form that decomposes information integration, adaptive evaluation and response-action formation into three mechanisms. The decomposition does not require three independent physical modules.

The standard type of Priority is $\mathcal{P}_{\mathrm{prio}} := \{\text{Priority 1}, \text{Priority 2}, \text{Priority 3}, \text{None}\}$. The standard set of operating modes is $\mathsf{Mode} := \{3\mathrm{A}, 3\mathrm{B}, 3\mathrm{C}\}$. The type of the response-style output of M3 is $\mathsf{Style} := \{\text{3B-own}, \text{3B-rel}, \text{3B-comb}, \text{none}\}$. Their meanings are given in the treatment of M2 and M3. The conceptual generation dependence inside TIPA is $M_1 \to M_2 \to M_3$: a necessary

upstream output becomes available to a downstream mechanism.

Assumption (`ass:functional_concurrency`): functional concurrency of the three mechanisms. M_1, M_2, M_3 can operate functionally in parallel at one time while preserving the generation dependence above. $M_1 \to M_2 \to M_3$ expresses information-generation dependence and does not fix the three mechanisms to strictly sequential modules in physical time or in hardware. *Introduction status: S (structural). Strength: medium.* It relies on the reading of the operation of the three mechanisms in the definition of the internal-model structure (`def:internal_model`). It is not used as a premise of a proposition of §4.

Sending and receiving between mechanisms, and same-time description (`rem:tipa_bidirectional`). In the discrete-time standard form, only the direction $M_1 \to M_2 \to M_3$ is placed within one time. A relation in which outputs are mutually inputs within one time would need, to fix each time's values, the existence of a fixed point and either its uniqueness or a deterministic rule fixing one value; §4 does not place it. Influence in the reverse direction is expressed across times, as an input at the next time. The standard form has two routes:

1. **Through retained records** (`def:s4_carrier_contract`). M1 takes as input the record of the previous M1 output $\mathrm{S}_{M1,t-1}$, the state-estimate history, the previous Priority P_{t-1} and the record of the final effective action u_{t-1}, and may use the record of the set of purposes in forming the purpose-objecthood component. M2 may read records of its own past evaluations ($\rho_{\mathrm{eval,past}}$ of `def:m2_latent_instability`). M3 takes the previous set of purposes Pur_{t-1} as input. The previous operating mode mode_{t-1} is not provided in the standard form as an independent retained record or explicit input; where information originating in it is actually held in an existing input such as M_t, its involvement in the current mode judgment is not excluded.
2. **Through the state and the external environment state.** The effective action u_t changes the next state and external environment state through f and the transition of the action-response component; the change enters M1 through sensing.

Each route holds only to the extent that the record or state actually carries the content. No route from the output of M3 to M2 is placed (`def:m2_map`, `rem:m2_no_return_scope`). The local protective response of PSL-4 is a route not passing through the three mechanisms (`subsubsec:psl4`), so $M_1 \to M_2 \to M_3$ is not the route through which all signals of the ACS pass.

In an implementation the three mechanisms are read as each receiving information from its connections and continuing to output processed information. The numbers are an order of description; bidirectional sending and receiving between mechanisms can occur during operation. What the paper describes for each mechanism extends only to processing received information within that mechanism's framework and giving an output that fits the type declaration. Iteration inside one mechanism is condensed into that mechanism's time-indexed map (`rem:internal_steps_time`). Where an implementation goes back and forth between mechanisms within one time, the going and returning is not condensed into the standard-form map: the time step is taken finer and the return is expressed as an input at the next time. The choice of time step is a descriptive choice, and the paper does not claim that such a step exists in every realization. Changing the step changes the one-step transition specification, and the judgment of the feasible action set and of blockage is relative to it (`rem:blockage_scope_sec2` (c)). Correspondence with continuous-time description is a Phase 2 task (L1 of `subsec:limitations_list`).

Scope of parallel processing of several items (`rem:multi_item_concurrency`). The discrimination state may contain several internal-object tokens, the request signals and request-by-object evaluations several requests and pairs, and the set of purposes several purposes. The generation dependence $M_1 \to M_2 \to M_3$ contains no rule that makes completion of processing of one item a condition for processing another. This paper therefore does not state, on the standard form, from one item being unprocessed, the stoppage of processing of other items as a general consequence. In an implementation several items can be in different stages at once. On the other hand, processing of individual items may be delayed or left unprocessed through the processing capability or specification of the implementation; on the standard form this appears as the map of the mechanism not reflecting the item in its output (for example, not assigning a token, or not including a request-by-object pair in the domain of evaluation). This paper does not state as a general consequence that all items are eventually processed,

that there is an upper bound on the delay, or fairness of processing. Unassigned tokens, unevaluated request-by-object pairs and withheld state judgments of purposes are quantities of different levels and are not merged into one unprocessed state.

Definition (`def:s4_output_attribution`): attribution of mechanism outputs to an individual (classification B). The time-indexed outputs of M1, M2, M3 are defined as outputs of Σ_i for times t with $\mathrm{WF}_{i,t} = 1$ (`def:wf_realization`). For $\mathrm{WF}_{i,t} = 0$ they are not defined as outputs of Σ_i; the output then does not take a particular value (0, `und`, the empty set or others) but is undefined. Times after collapse are included (the attribution provision of `def:collapse`). Each mechanism's map is total on its type; this provision concerns not the domain of the map but whether the time-indexed value is attributed to the individual. The functional axis (`def:functional_axis`) and other classification-C descriptive quantities are not its object.

Discrete time and internal processing steps (`rem:internal_steps_time`). $M_1 \to M_2 \to M_3$ and the standard map at each time do not express individual computations, neural processing, iterative comparison, re-evaluation or confirmation operations of an implementation one-to-one. The formation of one time's output can contain many internal steps, and the span from forming a purpose to judging its pursuit and attainment can cover several times. The paper does not stipulate the number of internal steps, the processing time or the number of iterations, nor a minimum. Iteration, re-comparison and re-integration inside a mechanism are condensed into its time-indexed map; going back and forth between mechanisms within one time is not condensed (`rem:tipa_bidirectional`). Whether an ACS holds long-term purposes, or purposes whose attainment requires persistence over a period, depends on the implementation.

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§4.2.1 M1: the information-integration mechanism

`Reference: subsec:m1`

M1 integrates the sensed information made available through PSL-1, the internal model, the finite history, the evaluation and action information of the previous time, and others, and constructs the internal representations of the current state and external environment state that are available to M2 and later. Implementation-dependent positive integers $W, W_{\mathrm{short}} \in \mathbb{N}$ with $1 \le W_{\mathrm{short}} \le W$ denote finite history lengths; $X^{\le W}$ is the set of X-valued finite histories of length at most W and $\mathcal{J}^{\le W_{\mathrm{short}}}$ the corresponding set for $\mathcal{J}$. A non-empty set Ξ is the discrimination state space made concrete in M1-b, and the standard simplex Δ^{n-1} is the range of the dynamics weights.

Definition (`def:m1_state`): M1 internal state (classification B). $\mathsf{S}_{M1,t} := (\hat{x}_t, \hat{e}_t, \xi_t, w_t, D_{M1,t})$ of type $\mathsf{S}_{M1} := X \times E_{\mathrm{ext}} \times \Xi \times \Delta^{n-1} \times \mathcal{D}_{M1}$. It bundles the already defined outputs of M1 as a standard internal representation and introduces no independent additional state variable or separate mechanism. Reference to $\mathsf{S}_{M1,t-1}$ by M1 at the next time is allowed; learning, plastic change and other additional inter-time update rules are left to §5.

Definition (`def:m1_map`): information-integration mechanism M_1 (classification B). With $\mathcal{J}$ the sensed-information space, $\mathcal{M}$ the internal-model state space, S_{M1} the M1 internal-state space and $\mathcal{D}_{M1}$ the space of time features for M2, M1 is a deterministic standard functional form

$$\mathrm{M1} : \mathcal{J} \times \mathcal{M} \times \mathcal{P}_{\mathrm{prio}} \times U \times X^{\le W} \times \mathsf{S}_{M1} \to X \times E_{\mathrm{ext}} \times \Xi \times \Delta^{n-1} \times \mathcal{D}_{M1},$$

with $(\hat{x}_t, \hat{e}_t, \xi_t, w_t, D_{M1,t}) = \mathrm{M1}(I_{\mathrm{sense},t}, M_t, P_{t-1}, u_{t-1}, \{\hat{x}_s\}_{s \in [\max(0,t-W),t-1]}, \mathsf{S}_{M1,t-1})$. The previous-time input u_{t-1} is a retained record of the final effective action and is distinguished from the record of an M3 action command; agreement with the actual previous action is a separate condition. At $t = 0$ the previous-time symbols take the readout from M_0^{W} (`def:s4_carrier_contract`); this does not assume a realization at a negative time.

M1-a: information integration

M1-a integrates currently available information and temporal context to form mechanism-level estimates of the current state and external environment state. As in §2, the realized external environment state and the M1 environment estimate are $e_t, \hat{e}_t \in E_{\text{ext}}$; the set E of required environmental factors is not used as the type of an external environment state.

Definition (`def:std_init`): preliminary state estimate of M1-a (classification B). $\Phi_{\text{pre}} : \mathcal{J} \times \mathcal{M} \times \mathsf{S}_{M1} \to X$, with $\tilde{x}_t := \Phi_{\text{pre}}(I_{\text{sense},t}, M_t, \mathsf{S}_{M1,t-1})$.

The external environment state estimate may be $\hat{e}_t = \Phi_{\text{env}}(I_{\text{sense},t}, M_t, \{\hat{x}_s\})$ with $\Phi_{\text{env}} : \mathcal{J} \times \mathcal{M} \times X^{\leq W} \to E_{\text{ext}}$.

Definition (`def:m1_integration`): dynamics weight and final integration of M1-a (classification B). With the dynamics weight $w_t \in \Delta^{n-1}$ and a final integration map $\Phi_{\text{integrate}} : X \times \mathcal{M} \times X \times \Delta^{n-1} \to X$, $\hat{x}_t = \Phi_{\text{integrate}}(\tilde{x}_t, M_t, \bar{x}_{\text{base},t}, w_t) \in X$.

Definition (`def:s4_m1_sources`): sources of the auxiliary outputs of M1 (classification B). The auxiliary quantities used from the preliminary estimate to the final integration are given by fixed maps $\Phi_{\text{weight}} : \mathcal{M} \times \mathcal{J} \to \Delta^{n-1}$ and $\Phi_{\text{baseline}} : \mathcal{M} \times \mathcal{J} \to X$, with $w_t = \Phi_{\text{weight}}(M_t, I_t)$ and $\bar{x}_{\text{base},t} = \Phi_{\text{baseline}}(M_t, I_t)$. Φ_{baseline} is the composite that applies the history aggregation map Φ_{hist} (`def:m1_temporal_base`) to the state-estimate history (third component) read out from the working component of M_t by ρ_{past}. If a one-step prediction is used, a fixed $\Phi_{\text{predict}} : \mathcal{M} \to X^{X \times U \times E_{\text{ext}}}$ gives $\hat{f}_t = \Phi_{\text{predict}}(M_t)$; the exponent denotes the set of all functions from $X \times U \times E_{\text{ext}}$ to X, and continuity is not required. The previous estimate used in the prediction is also taken from the retained record.

State and environment estimates versus realized values (`rem:m1_estimate_nontruth`). $\hat{x}_t$ and $\hat{e}_t$ are internal estimates; agreement with x_t and e_t is not required in general, and §4 does not require a universal bound on estimation error or convergence.

Definition (`def:s4_state_estimation_error`): state estimation error (classification C). For times with $\text{WF}_{i,t} = 1$, with $X \subseteq \mathbb{R}^n$ and the restricted distance $d_X(x, x') = \|x - x'\|_\infty$, $\varepsilon_{\text{est},t} := d_X(x_t, \hat{x}_t)$. It is a quantity compared with the true state and is distinct from the internal prediction-consistency error $\varepsilon_{\text{pred},t}$. Internal computability or external measurability is not required.

M1-b: boundary management

M1-b constructs, by several abstract discrimination axes, the discrimination states needed for response formation, including the self–environment boundary.

Definition (`def:id_vector`): discrimination state and internal-object tokens (classification B). The index set of discrimination axes is $\mathcal{R}_\xi := \{\text{bnd}, \text{tmp}, \text{org}, \text{dyn}, \text{rep}, \text{mval}, \text{prox}, \text{tgt}\}$. For each $\kappa \in \mathcal{R}_\xi$ a finite non-empty set Ξ_κ containing an undecided category is placed, and the range of one row is $\Xi^{(8)} := \prod_{\kappa \in \mathcal{R}_\xi} \Xi_\kappa$. To keep the type interface upstream, the boundary-attribution axis is $\Xi_{\text{bnd}} = \{\text{self}, \text{env}, \text{und}\}$ (attribution to the self side, to the environment side, undecided); the concrete category decomposition of the other seven axes is not fixed in general in Phase 1.

A finite non-empty set Lab of labels of internal-object tokens is fixed. The symbol $*$ (used to index the overall row) and self do not belong to Lab, and $\text{Lab} \cap (\{\text{self}\} \times \text{Lab}) = \emptyset$. Its size depends on the implementation. The discrimination state space is $\Xi := \Xi^{(8)} \times \bigcup_{\mathcal{K} \subseteq \text{Lab}} (\Xi^{(8)})^{\mathcal{K}}$, and the discrimination state of M1-b is $\xi_t = (\xi_t(*), (\xi_t(k))_{k \in \mathcal{K}_t}) \in \Xi$. The first component $\xi_t(*) \in \Xi^{(8)}$ is the *overall row*, the index set $\mathcal{K}_t \subseteq \text{Lab}$ of the second component the *internal-object token set* at time t, and each $\xi_t(k) \in \Xi^{(8)}$ the *row* of token k. $\mathcal{K}_t = \emptyset$ is allowed. The κ-th components are written $\xi_{\kappa,t}$ and $\xi_{\kappa,t}(k)$. In particular $\xi_{\text{bnd},t}$, referred to in §2, is the boundary-attribution component of the overall row.

The axes express boundary attribution, temporal position, information origin, dynamics, representation basis, benefit or harm in memory matching, interaction proximity and purpose-objecthood. Concrete categories, feature-extraction rules and internal coding depend on the

implementation.

Formation of tokens and the implementation-dependent range (`rem:token_generation`). $\mathcal{K}_t$ is part of the output ξ_t of M1 and can be formed at the same time from the inputs of M1 (the sensed information $I_{\mathrm{sense},t}$, the internal model M_t and the retained records). This formation does not use the same-time outputs of M2 and M3, so no same-time circularity arises. The paper does not require: (a) that tokens be assigned to all objects that exist externally; (b) that a newly appearing object be assigned a token at the time it is sensed; (c) that each component of a token's row agree with the properties of the object. The number of tokens that can be formed at one time, the delay and precision of assignment, and whether several objects are identified one by one or simultaneously depend on the implementation. For $\mathcal{K}_t = \emptyset$ the overall row $\xi_t(*)$ is still defined, and M2 and M3 can use it for evaluation and response formation. A label $k \in \mathsf{Lab}$ is an internal unit of discrimination and is not identified with a slot $a \in \mathcal{A}_O$ of §2 or an external individual; the same label may be reused across times for a different object. Persistence of a token across times is expressed as M1, taking $\mathsf{S}_{M1,t-1}$ as input, giving the same label; the rule of that persistence is left to §5.

The meanings of the axes are as follows.

- **Boundary attribution:** whether an object or information is treated as self side, environment side or undecided.
- **Temporal position:** whether it relates to present, past or undecided.
- **Information origin:** internal origin, external origin or undecided.
- **Dynamics:** whether an object is discriminated as a dynamic object for interaction, a non-dynamic object or undecided.
- **Representation basis:** whether a representation is of observational origin, internally generated origin or undecided.
- **Benefit or harm in memory matching:** the relation between current information and past experience in memory matching; it is not itself the current feasibility evaluation.
- **Interaction proximity:** the degree to which interaction with an object can hold for the current ACS; not identified with mere physical distance.
- **Purpose-objecthood:** whether, on the records held at the start of time t, an object or information is the direction of a purpose of the ACS, is not, or is undecided. It does not refer to the purpose Pur_t that M3 forms at the same time (`def:purpose_formation`). A token that newly becomes the direction of a purpose at time t can appear in this axis at $t+1$ and later through the records. Accuracy of the records is not required in general (`def:s4_carrier_contract`). M2 does not use this axis in evaluation (`def:m2_map`).

Separation of memory-matching benefit or harm from M2 evaluation (`rem:mval_identification`). This paper does not state, as a general consequence, the current Priority, feasibility threat or action request from similarity to an object that was beneficial or harmful in the past alone. The current feasibility-related evaluation is the responsibility of M2.

Role and non-truth of the discrimination state (`rem:id_vector_role`). ξ_t is a standard internal representation of M1. It is not the ontological self–environment boundary of the ACS, an objective attribute of an external object, or the judgment of a time-indexed satisfaction predicate of the three principles.

M1-c: integration of temporal continuity

M1-c uses the finite history and the prediction by the internal model to construct the temporal context and time features needed for present processing.

Definition (`def:m1_temporal_base`): temporal base representation of M1-c (classification B). With a history aggregation map $\Phi_{\mathrm{hist}} : X^{\leq W} \to X$, $\tilde{x}_{\mathrm{base},t} := \Phi_{\mathrm{hist}}(\{\hat{x}_s\}) \in X$.

If the internal model uses a one-step prediction, it is typed $\hat{f}_t : X \times U \times E_{\mathrm{ext}} \to X$; when the previous final effective action u_{t-1} and environment estimate $\hat{e}_{t-1}$ are used, $\tilde{x}_t^- := \hat{f}_t(\hat{x}_{t-1}, u_{t-1}, \hat{e}_{t-1})$.

Definition (`def:m1_temporal_features`): prediction consistency and leading time features of M1-c (classification C). With a distance d_X on the state space, the prediction-consistency error $\varepsilon_{\mathrm{pred},t} := d_X(\tilde{x}_t, \tilde{x}_t^-) \in \mathbb{R}_{\geq 0}$ may be described. These are mechanism-level descriptive quantities of classification C and do not express the correctness of a prediction or the truth of a time-indexed satisfaction predicate of the three principles.

Definition (`def:env_set_prediction`): set-valued prediction on the external side (classification B). Following the decomposition $E_{\mathrm{ext}} = E^{\mathrm{bg}} \times O$ (`def:Eext_decomposition`), the external environment state and its estimate are $e = (b, o)$ and $\hat{e}_t = (\hat{b}_t, \hat{o}_t)$. The set of external-side prediction values is $\mathcal{G}_O := (\mathcal{P}(O) \setminus \{\emptyset\}) \cup \{\bot\}$, where $\bot$ means that no prediction is given and is not the empty set. If the internal model uses a prediction of the action-response component, a fixed map $\Phi_{\mathrm{pred},O} : \mathcal{M} \to \mathcal{G}_O^{X \times U \times E_{\mathrm{ext}}}$ gives $\hat{g}_t := \Phi_{\mathrm{pred},O}(M_t)$, and the prediction set for an action $u \in U$ is $\hat{G}_t(u) := \hat{g}_t(\hat{x}_t, u, \hat{e}_t) \in \mathcal{G}_O$. $\hat{G}_t(u)$ is the set of values of the action-response component at the next time that the ACS internally anticipates as possible; it expresses neither probability nor frequency. $\hat{g}_t$ is a map distinct from the state-side one-step prediction $\hat{f}_t$.

Definition (`def:prediction_record_check`): record of the prediction set and two kinds of deviation from prediction (classification C). After the effective action u_t at time t is fixed, the prediction set $\hat{G}_t(u_t)$ at the time of acting is left in the retained records, and at $t+1$ its recorded value $\hat{G}_t^{\mathrm{rec}} \in \mathcal{G}_O$ is obtained from the readout of `def:s4_carrier_contract`; agreement of the recorded value with $\hat{G}_t(u_t)$ is not assumed. Two notions are defined:

- (i) *internal deviation from prediction*: when $\mathrm{WF}_{i,t+1} = 1$ and $\hat{G}_t^{\mathrm{rec}} \neq \bot$, that $\hat{o}_{t+1} \notin \hat{G}_t^{\mathrm{rec}}$. It is determined by quantities available to M1-c at $t+1$ and is a descriptive quantity at the same level as $\varepsilon_{\mathrm{pred},t}$ of `def:m1_temporal_features`.
- (ii) *theoretical deviation from prediction*: when $\mathrm{WF}_{i,t} = 1$ and $\hat{G}_t(u_t) \neq \bot$, that $o_{t+1} \notin \hat{G}_t(u_t)$. It is a quantity compared with realized values and is not an input of the ACS (treated like $\varepsilon_{\mathrm{est},t}$ of `def:s4_state_estimation_error`).

For each notion, if its respective prediction value is $\bot$, that notion is undefined rather than false. The undefinedness of one notion does not by itself make the other undefined. The two need not coincide; for example, when misrecognition of an object makes $\hat{o}_{t+1}$ differ from o_{t+1}, only one of them can hold.

Difference of the reference for the two predictions (`rem:prediction_reference_difference`). The state-side $\tilde{x}_t^-$ is the value recomputed at time t by the prediction map $\hat{f}_t$ from the previous-time record. On the external side, by contrast, the comparison is with the record of the prediction set at the time of acting. If the internal model was updated between t and $t+1$, the recomputed prediction and the prediction at the time of acting need not coincide. The paper requires neither replacing the former by the latter nor the converse. Rules for updating the internal model on the basis of a deviation from prediction are not given here (§5).

Definition (`def:m1_temporal_output`): final time features of M1-c (classification B). The time features that M1-c provides to M2 are $D_{M1,t} \in \mathcal{D}_{M1}$. The internal components of $\mathcal{D}_{M1}$ may include state change, input change, prediction difference, trend of change and others, depending on the implementation.

Generation dependence inside M1. Conceptually: current input and history → preliminary estimate and temporal context → final integration → discrimination and time features. This does not require physical sequential execution.

Assumption (`ass:m1c_temporal`): temporal integration structure of M1-c. M1-c can connect the finite history to present processing and form time features available to M2. *Introduction status: S (structural). Strength: medium.* It limits the range of ACSs to which the standard form of M1-c (`def:m1_temporal_base`, `def:m1_temporal_output`) applies, and is not used as a premise of a proposition of §4.

Responsibility boundary of M1 (`rem:m1_responsibility`). M1 bears state and environment estimation, discrimination and temporal integration. It does not bear Priority generation, generation of request signals, formation of purposes, judgment of the operating mode, comparison of action candidates, or judgment of the time-indexed satisfaction predicates of the three principles.

Check status. This summary adds no new definition, assumption or result. Labels in backticks are shared with the manuscript. It has been checked against the manuscript by the drafting model and audited by an external AI model. It remains a draft until the author checks it against the manuscript. No independent human third-party verification is claimed.